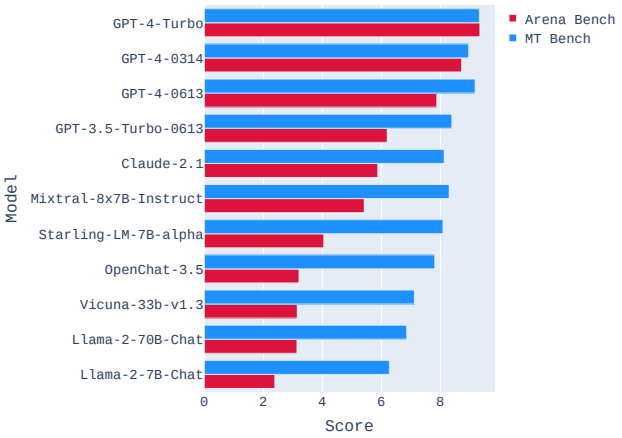

Figure 4. Model's performance between Arena Bench and MT-Bench, showing an increased gap between open and proprietary models. Both uses GPT-4 as judge.

Table 3. Pairwise agreement rate between crowd-user, gpt-4 judge, and experts on pairwise battles. The top part of the table is between GPT-4-Turbo and Llama-2-13b-chat. The bottom is between GPT-4-Turbo and GPT-3.5-Turbo-0613.

Llama-2-13b	Expert 1	Expert 2	GPT-4
Crowd	72.8%	77.8%	75.6%
Expert 1	-	89.8%	81.0%
Expert 2	-	-	78.5%
GPT-3.5-Turbo	**Expert 1**	**Expert 2**	**GPT-4**
Crowd	73.8%	83.1%	75.6%
Expert 1	-	79.4%	76.3%
Expert 2	-	-	79.3%

data point took on average 3-5 minutes. For reference, we also use GPT-4 as a judge for pairwise comparisons. The agreement rate between crowd-users, experts, and GPT-4-judge are presented in Table 3. The corresponsing win-rate are shown in Table 4.

To summarize, we observe high agreement rates (72% to 83%) between Arena crowd-user and experts in both setup. Note that agreement rates between two experts are around similar levels (79.4% and 89.8%). As for the 10%-20% disagreement between experts, it is mostly due to some user prompts don't have a ground truth answer. Depending on the preference of the evaluator, sometimes both answers can be argued as being better than the other one, such as the examples in Appendix D.4. The gap between crowd-vs-expert agreement rate and expert-vs-expert agreement rate (5%-10%) is mostly attributed to crowd user making mistakes or overlooking factual errors in model's response. Overall, the agreement rates presented in Table 3 validate the decent quality of crowd-sourced votes in Chatbot Arena.

Table 4. GPT-4-Turbo's win-rate across crowd-user, gpt-4 judge, and experts on pairwise battles against Llama-2-13b and GPT-3.5-Turbo-0613.

Baseline	Arena User	Expert 1	Expert 2	GPT-4
Llama-2-13b	81.2%	89.4%	86.9%	78.8%
GPT-3.5-Turbo	76.3%	82.5%	89.4%	79.4%

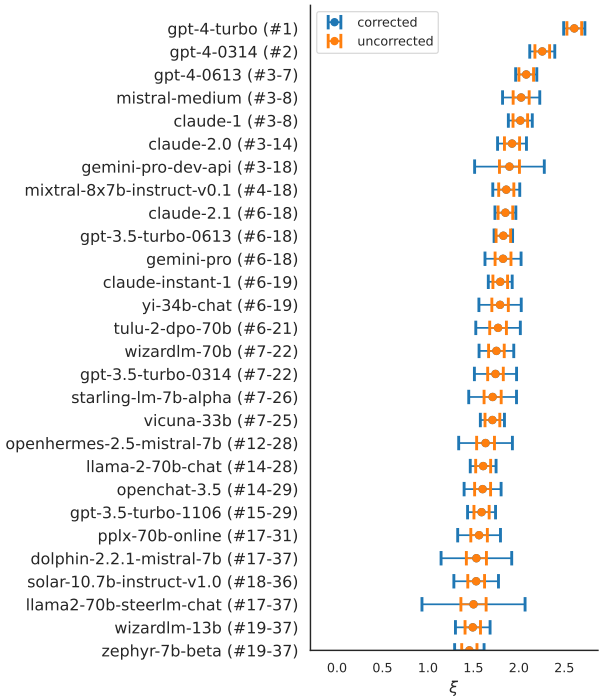

Figure 5. Intervals for the BT coefficients with and without multiplicity correction. The multiplicity correction, in this case a chi-square CLT interval, is technically required for the purpose of calculating the ranking, because it ensures *all* scores are simultaneously contained in their intervals (and the ranking is a function of all the scores). However, it induces extra conservatism, so we report both intervals.

7. Experiments

7.1. Ranking system

Computing the rank on real data. In this section, we report results from our experiments on approximate ranking. For this experiment, we ran a replay of $T = 213,576$ historical votes from our online platform and calculate the BT coefficients using our earlier-described estimation algorithm with confidence intervals; see Figure 5 for these intervals (with and without multiplicity correction; the formal notion of approximate ranking technically requires multiplicity correction, but it makes the intervals looser).

Evaluating the coverage of the intervals. A natural follow-up question is whether or not the intervals are doing their job correctly: whether they cover the true BT coefficients with probability at least (and almost exactly) $1 - \alpha$. Of course,

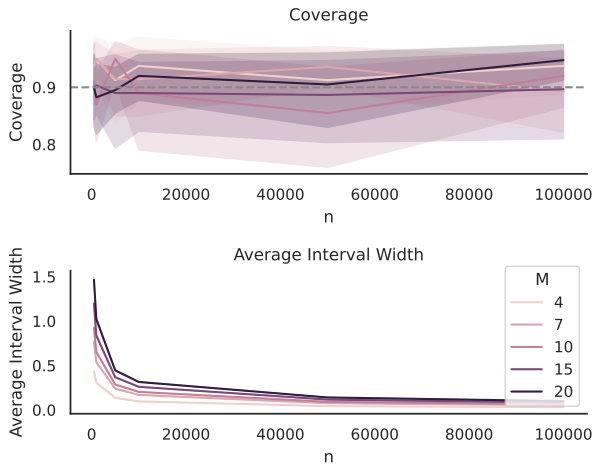

Figure 6. Intervals for the BT coefficients as a function of the number of samples and the number of models M.

this cannot be evaluated on real data, so we run a simulation. A vector of BT coefficients is drawn, with each coordinate sampled i.i.d. from a distribution $\texttt{beta}(1/\gamma, 1/\gamma)$; we take $\gamma = 2$ in Figure 6 (and we vary γ in Appendix A). Given these coefficients, a dataset is synthesized, and the coverage and average width are computed for each of 20 trials. The results can be seen in Figure 6 for the uncorrected intervals The coverage of the intervals behaves as expected, centering around $1 - \alpha$, regardless of the number of models. Meanwhile, the more models are included, the larger the intervals become.

Evaluating the active sampling rule. Next, we discuss the evaluation of our active sampling rule as Equation (9) for win matrix estimation. We evaluate this sampling rule by taking the best fit BT coefficients to our 213,576 point sized holdout set, and then sampling from that distribution using our active sampling algorithm. The results are displayed in Figure 7. It is hard to tell by looking at plots, but the improvement is substantial: To estimate θ^* to a precision of 0.2, random needs 6,800 samples and adaptive needs 4,400 samples; meanwhile to estimate the score to a precision of 0.3, random needs 17,200 samples and adaptive needs 16,400 samples. Thus, the random baseline requires 54% and 5% more data to achieve the same level of precision, respectively. One can see from the plots in Figure 7 that these results are not cherry-picked: the sample-efficiency of our method is better at all values on the horizontal axis.

7.2. Anomalous Users Detection

We evaluate the outlier detection method in Section 5.1. We construct the evaluation set by manually identifying 25 anomalous users whose inputs are highly repetitive or meaningless (e.g., asking "hi" for 100 times or inputting garbled texts). We randomly sample 25 normal users with

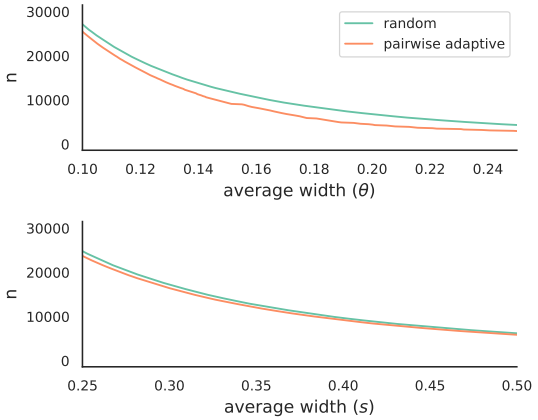

Figure 7. Interval widths on the win matrix (upper figure) and on the BT coefficients (lower figure) as a function of the number of samples, for random sampling and also adaptive sampling. Improvements from adaptive sampling can be seen in both cases, although they are more subtle on the scale of the score.

Table 5. Confusion matrix of different α. "Pred." means predicted. Positive means anomalous and negative means normal.

$\alpha = 0.1$	Pred. Positive	Pred. Negative
Actual Positive	13/14	12/36
Actual Negative	1/14	24/36
$\alpha = 0.3$	Pred. Positive	Pred. Negative
Actual Positive	21/29	4/21
Actual Negative	8/29	17/21

at least 50 votes, and inspect their input prompts to ensure no abnormal behaviors. As mentioned in Section 5.1, per user we compute five M_j and identify the user as anomalous if $M_j \geq \chi^2_{2j,1-\alpha/5}$. We present results of two different α (i.e., the significance leval) in Table 5. We find the detection method effective (e.g., reaching 90% true positive and 60-70% true negative rate). We inspect the false negative errors and find those are from users do not always behave abnormally, making them harder to detect.

8. Discussion

Limitations. Although our user base is extensive, we anticipate that it will primarily consist of LLM hobbyists and researchers who are eager to experiment with and evaluate the latest LLMs. This inclination may result in a biased distribution of users. Additionally, despite the wide array of topics encompassed by the prompts discussed in previous sections, the data predominantly comes from our online chat interface. This source might not accurately reflect the real-world usage of LLMs in production environments or specialized domains, potentially leading to a skewed prompt distribution. Moreover, our study concentrates on assessing the helpfulness of LLMs but overlooks their safety aspects. We recognize the possibility and necessity of a parallel

mechanism to evaluate the safety of these models.

Future Directions. In our future work, we plan to develop comprehensive topic leaderboards and establish a dedicated section for multimodal and agent-based LLMs in more dynamic, gamified settings, catering to more complex tasks. We also believe our approach to detecting harmful users could be improved and made more formally rigorous by using the theory of nonnegative supermartingales and E-values (Howard et al., 2020; Waudby-Smith & Ramdas, 2020; Vovk & Wang, 2021; Ramdas et al., 2023); this would deal with the dependence, but the variants we tried did not perform well in terms of power.

9. Conclusion

In this paper, we present Chatbot Arena, an open platform for evaluating LLMs through crowdsourced, pairwise human preferences. We conduct an in-depth analysis of the crowdsourced user prompts and preference votes to validate the diversity and quality. We develop an efficient model sampling and ranking algorithm. Our dataset including 100K pairwise preference votes will be released for future research.

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